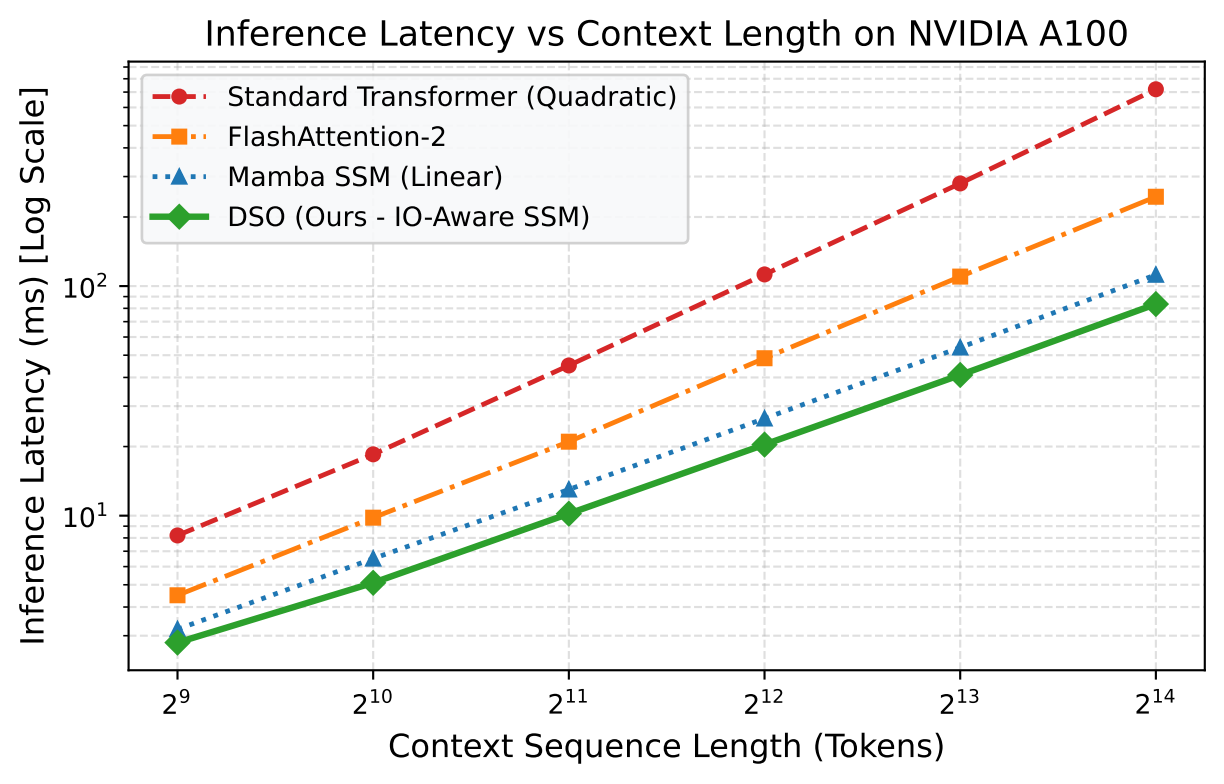
Inference Latency vs Context Length on NVIDIA A100
Standard Transformer (Quadratic)
FlashAttention-2
Mamba SSM (Linear)
DSO (Ours - IO-Aware SSM)
Inference Latency (ms) [Log Scale]
10^2
10^1
2^9
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
Context Sequence Length (Tokens)